Electronic Mail Architecture & Protocols

Coverage: User Agents (UA), Mail Transfer Agents (MTA), SMTP Push Protocol (Port 25/587), MIME Base64 Encoding, POP3 vs IMAP4 vs Webmail

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. Email Delivery Flow & SMTP

- SMTP (RFC 5321): Push protocol over TCP port 25 (server-to-server) and port 587 (client submission).
- Delivery: Sender UA -> Sender MTA (SMTP) -> Receiver MTA (SMTP) -> Receiver UA (POP3/IMAP4/Webmail).

2. MIME & Access Protocols

- MIME: Enables binary attachments over 7-bit ASCII SMTP using Base64 encoding.
- POP3 (Port 110): Download-and-delete mode. IMAP4 (Port 143): Synchronized server folders and partial message fetches.

■ KTU University High-Yield Exam Questions

- Q: Differentiate between SMTP and POP3/IMAP4. Explain why POP3/IMAP4 are required for email access. (7 Marks)